



Static and dynamic properties of a perforated metallic auxetic metamaterial with tunable stiffness and energy absorption

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ABSTRACT

Auxetic metamaterials have attracted increasing attention due to their exceptional mechanical properties. However, the critical parameters of mechanical response and Poisson's ratio would be changed simultaneously when a geometrical parameter is tuned, which is adverse to achieving the quantitative design of energy absorption by tuning a single geometrical parameter. Thus, the methodology based on tuning densification strain is proposed to design auxetic unit cells with tunable stiffness. In this study, the static performance of 2D metallic auxetic metamaterials designed by the variable stiffness factor (VSF) method is examined experimentally and numerically. To further achieve tunable energy absorption under crushing load, the concept of VSF is extended to variable energy factor (VEF). The dynamic response of verified numerical models is investigated subjected to low-, medium-, and high-velocity crushing. Finally, a functionally graded auxetic structure with different design layers is proposed to effectively solve the issues of high stiffness ratio and initial peak force. These results show that the designed structure has the actual VSF and VEF percentages close to the designed value under low- and medium-velocity crushing. The findings from this study are useful for wider applications of auxetics in protective engineering.

1. Introduction

During the past decades, materials with negative Poisson's ratio have attracted growing attention due to their counterintuitive deformation, i. e., they could expand (shrink) laterally under tension (compression) [1–4]. Along with this special characteristic, negative Poisson's ratio materials are proved to have many desirable properties, e.g., shear resistance [5], fracture resistance [6], indentation resistance [7] and energy absorption [8–10], etc. Therefore, many potential applications of auxetics were reported in a wide range of fields including base seismic isolation and dampers in earthquake engineering [11,12], buffer devices in traffic engineering and aerospace [13,14], sensors in instrumentation science [15] and vascular stent in the medical area [16,17].

In 1987, the first artificial negative Poisson's ratio foam was fabricated and reported in the seminal work of Lakes [18], which inspired the enthusiasm of researchers to develop more structural forms with negative Poisson's ratio effect. To avoid the cumbersome phrase, “negative Poisson's ratio materials” was named “auxetics” by Evans [19,20].

Several cellular auxetic materials have been proposed in the last decades, including re-entrant structures [21], chiral structures [22], rotating polygonal structures [23], perforated sheets structures [24] and other structures [25,26].

However, the aforementioned auxetic metamaterials have low in-plane stiffness due to the high porosity, which hindered their engineering applications. To invent a novel auxetic structure with enhanced stiffness, many scholars strive to modify the geometries of original auxetic unit cells [27,28]. For re-entrant honeycombs, merging two perpendicular unit cells into a hybrid honeycomb [29], embedding rhombus into the interior of concave hexagon [30], and adding a strengthening rib along the re-entrant direction [31] could effectively improve the stiffness (Fig. 1(a)). Besides, composites composed of soft materials (matrix) and auxetic frames (reinforcements) are also widely studied in recent years [32–34], (Fig. 1(b)). The stability and stiffness of such composites could be significantly increased compared with pure auxetic frames [35].

With the development of auxetics, achieving tunable auxeticity and

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stiffness has become a new goal. Li et al. [36] reported a novel 3D auxetic unit cell with butterfly-like perforation, which exhibited tunable stiffness by designing its different geometric parameters. Although the tunable range of stiffness is wide, tuning the stiffness ratio quantitatively by the geometric characteristics is still a puzzle. For perforated sheets structures, the mechanism of auxetic behavior is the rotation of unit cells [37–39]. Mizzi et al. [40] reported a highly tunable auxetic metamaterial with diamond and elliptical perforations to exhibit a large range of Poisson's ratio by altering the geometric parameters and orientation of perforations. Yao et al. [41] proposed a graded perforated auxetic plate, which exhibited a higher degree of compliance of overall stiffness. Ren et al. [42–44] proposed an improved methodology to generate 3D buckling-induced auxetic metamaterials using one single control parameter of pattern scale factor (PSF). The Poisson's ratio and stiffness of generated auxetic metamaterials could be simply tuned by PSF. According to these publications, auxetic metamaterials usually possess a densification stage where the compressed stiffness increases sharply [45,46]. Thus, from the aspect of tuning densification strain, a novel methodology was firstly reported to design auxetic perforated structures to achieve tunable stiffness indirectly [47]. One of the primary properties of tunable auxetic metamaterials is energy absorption capacity. Therefore, examining the structural mechanical properties of tunable stiffness under crushing load becomes important.

The structural deformation mechanism and load vs displacement response are significant to estimate the energy absorption performance. Due to the high compressibility and porosity, cellular materials tend to possess a long stress plateau and densification stage under axial crushing load, which is preferable for the application of anti-impact, e.g., honeycomb, foam, wood [48]. Especially, hexagonal honeycombs [49,50], graded cellular materials [51,52], and octet-truss structures [53,54] have been widely studied recently. However, they usually exhibit higher peak force and more jagged stress plateau than the corresponding buckling-induced cellular materials, i.e., auxetics [55–58] and origami [59,60]. Thus, the energy absorption performance of auxetic materials, including re-entrant hexagonal honeycomb [61,62], double arrowhead honeycombs [63–65], chiral cellular structures [66], and oval structures [67,68], has been investigated extensively. But their critical parameters of mechanical response, e.g., the peak force, plateau stress densification strain and Poisson's ratio would be changed simultaneously when a geometrical parameter (wall thickness, concave angle, or ellipticity, etc.) is tuned. This is adverse to achieving the quantitative design of energy absorption by tuning a single geometrical parameter.

In this work, based on the variable stiffness factor (VSF) method [47], a series of metallic auxetic metamaterials were manufactured and tested under axial quasi-static compression for examining the accuracy of tunable stiffness. By tuning a single VSF value, the quantitative design of energy absorption is explored. Inspired by the VSF method, an extended indicator, variable energy factor (VEF) was proposed to evaluate the accuracy of tunable energy absorption under crushing load numerically. Besides, a novel functionally graded auxetic structure was designed and manufactured to decrease the equivalent modulus in densification points. The aim of this research is to achieve a structure with tunable and controlled energy absorption according to the proposed design methodology, meeting the requirements in practical applications.

2. Designing the auxetic metamaterial with tunable stiffness

The methodology of designing the auxetic metamaterial with tunable stiffness can be divided into four steps. Firstly, generating the original buckling-induced auxetic unit cell using the PSF method [43,45,46]. The design regions and the variation range of VSF would be reduced when the original ellipse becomes narrow and oblate. Thus, the elliptical ratio of short-axis to long-axis 0.8 (PSF=20%) was adopted as the original one in this paper for enough design regions. The range of VSF percentage is 36.3%~100% under this original auxetic unit cell. As shown in Fig. 2(a), the designed unit cell was divided into plastic deformation region (gray), design region of variable stiffness (purple), and optimized region (green). Secondly, building the unit cell of VSF=100% by adding materials in the design area. The red point represents the center of the rotation in Fig. 2(a), which is defined as the point of intersection between the cutting line and the lower boundary of the design region. Along the cutting line, the unit cell with VSF=100% is generated based on the rotating mechanism. The geometric sizes in Fig. 2(b) are designed as $a = 5 \text{ mm}$, $b = 4 \text{ mm}$, $c = 1.5 \text{ mm}$, $h = 1 \text{ mm}$. Thirdly, conducting rotary cutting to generate auxetic unit cells with required VSF percentages, i.e., VSF = 40%, 60%, and 80% in Fig. 2(c). The variable stiffness factor η_d (VSF) is subjected to rotating angle θ_l and defined as Eqs. (1) and (2) [47]. Lastly, arraying the unit cell along with the cylindrical or orthogonal direction to form the required auxetic metamaterials, as shown in Fig. 2(d). The specific design has been demonstrated in our previous work [47].

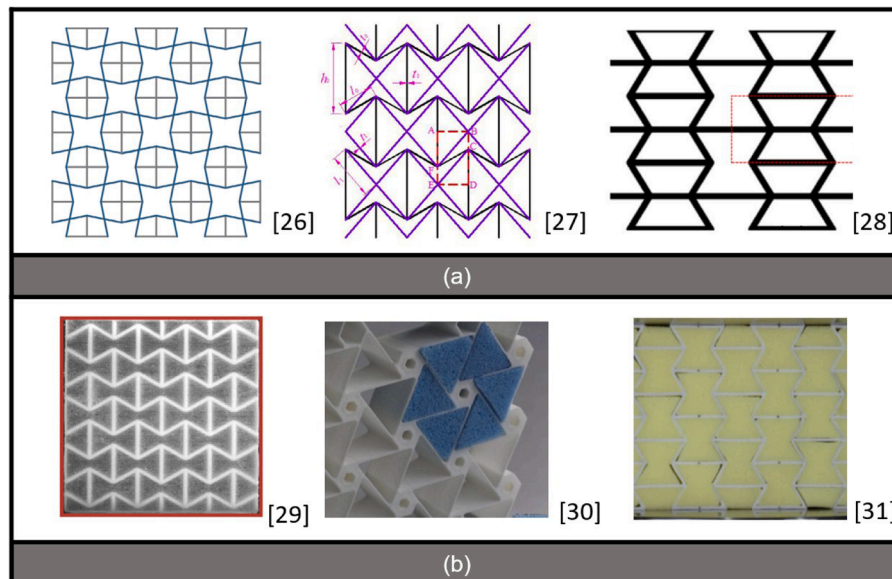


Fig. 1. Auxetic materials with enhanced stiffness.

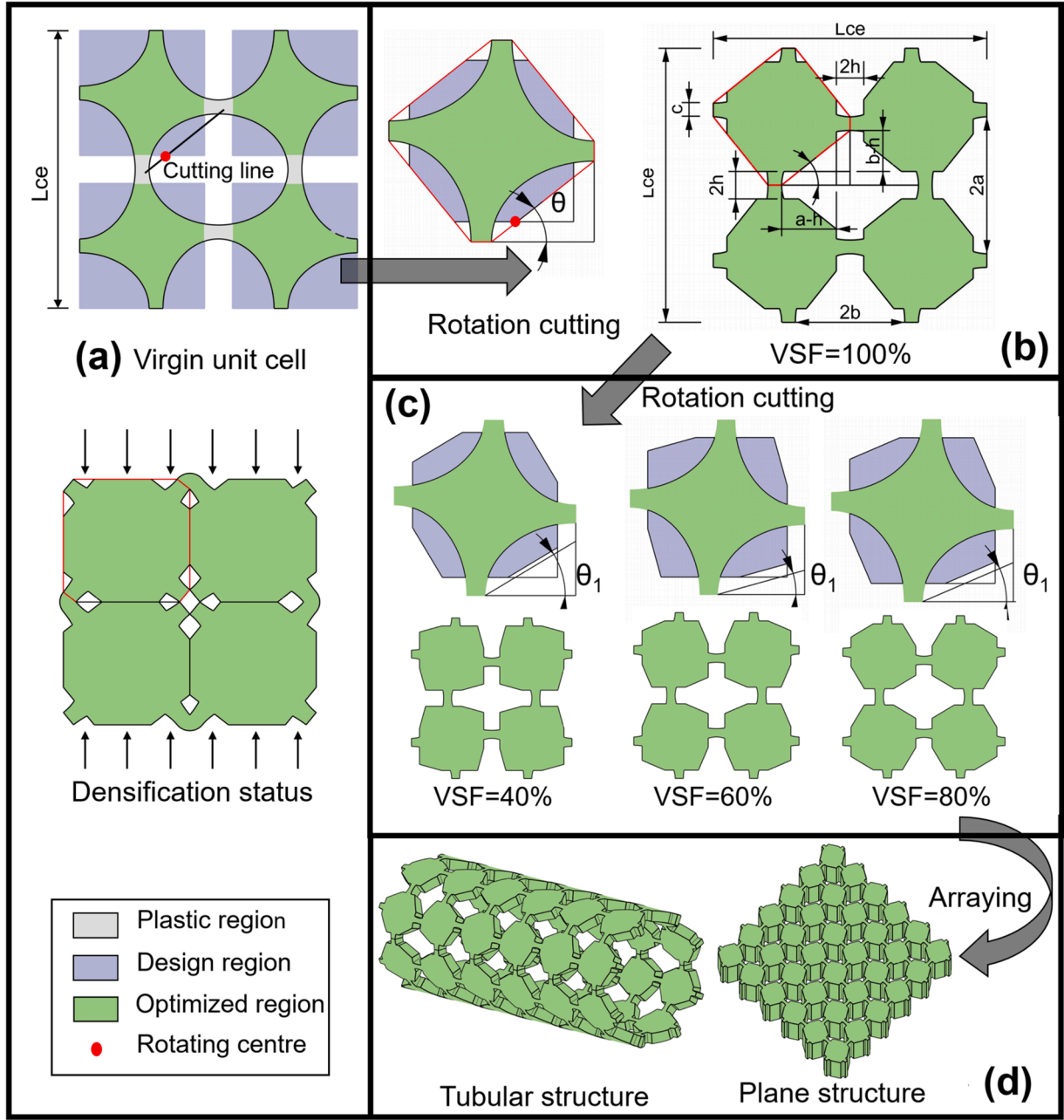


Fig. 2. The design process of auxetic metamaterials with tunable stiffness: (a) original unit cell, (b) structural design of VSF = 100%, (c) unit cells with different VSF percentages, (d) tubular and plane structures.

$$\eta_d = \frac{\theta_1}{\theta}$$

$$\frac{\arctan\left(\frac{h}{a-h}\right)}{\arctan\left(\frac{b}{a}\right)} \leq \eta_d \leq 1$$

where η_d is the designed proportion of VSF, θ_1 is the angle between the cutting line and the horizontal line in the designed model, θ is the angle between the cutting line and the horizontal line in the model of VSF = 100%.

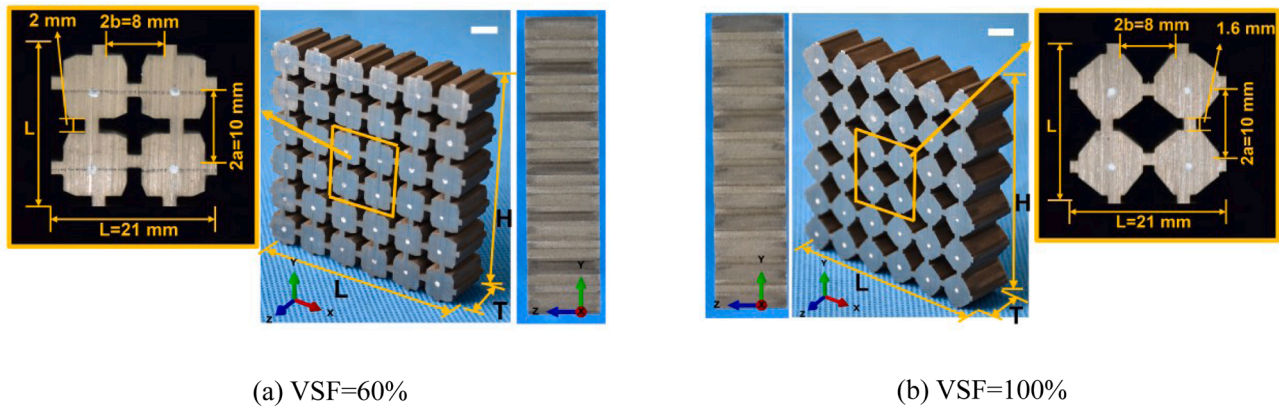
3. Experimental verification

3.1. Fabrication of the specimens

The specimens were fabricated by perforating designed holes in a solid stainless steel plate using a laser cutting machine. The fabricated specimens with VSF = 60% and VSF = 100% were shown in Fig. 3. Similar to our previous work [47], the dimensions of two samples with same $L \times T \times H$ of $63 \times 15.5 \times 63 \text{ mm}^3$.

3.2. Material property test of stainless steel

Five dog-bone specimens possessed the same type of stainless steel with mentioned compressed specimen are cut according to the standard [69]. The uniaxial tensile test was performed with a loading speed of 2



(a) VSF=60%

(b) VSF=100%

Fig. 3. The specimens fabricated using laser cutting (Scale bar: 10 mm).

mm/min using a universal machine (WANCE ETM-104B, China), as shown in Fig. 4(a). The relevant material properties were recorded in Table 1.

3.3. Uniaxial compression tests

The mechanical response of the auxetic metamaterials was tested using a universal testing machine (SUNS WAW-600, China) with a strain rate of 10^{-3} 1/s in Fig. 4(b). A camera was used to record the deformation in lateral and uniaxial directions so as to calculate the evolution of Poisson's ratio by image processing discussed in the following section [45]. Due to the existence of the densification stage for auxetic materials, the reaction force in the plateau stage is relatively smaller than that after densification. To guarantee the accuracy of data before densification, the testing machine with a loading range of 600 kN was employed and the compression process was stopped manually when the force is close to 600 kN.

3.4. Finite element analysis

To validate the effectiveness of experiment and parameter analyses, finite element analysis was conducted using Abaqus/Explicit solver. The geometric large deformation and complex contact nonlinearity were taken into consideration [70]. For the interaction module in the software, the normal behavior of hard contact and tangential behavior of the penalty function was chosen in general contact. The friction coefficient was set as 0.15. The models were established using solid element (C3D8R) with the global size of 0.4 mm, under which all ligaments have been divided into at least three layers of mesh for avoiding shear locking phenomena. The verification of mesh density and loading rate was conducted according to our previous work [42]. Two rigid plates were

Table 1

Material properties of stainless steel.

Material	Thickness t (mm)	Young's modulus E (GPa)	Yield stress σ_y (MPa)	Ultimate strength σ_u (MPa)	Ultimate strain ϵ_u
stainless steel 304	1	190	360	863	0.158

employed to simulate the squeeze heads in the test (Fig. 5). A bilinear elastoplastic constitutive model was employed in finite element models. The specific material properties have been given in Table 1.

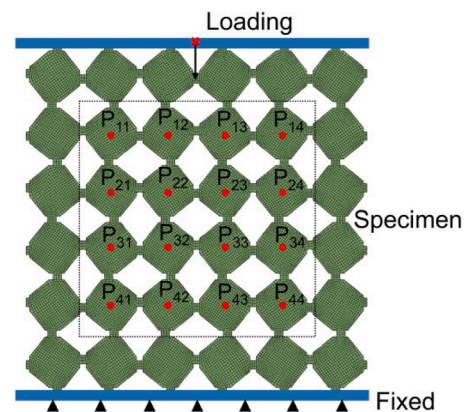


Fig. 5. The schematic diagram of finite element model.

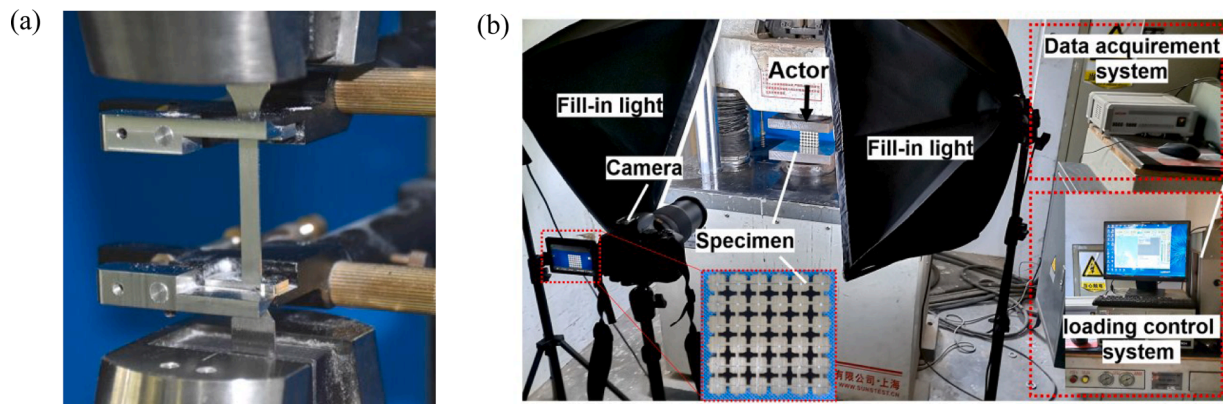


Fig. 4. Experimental setups: (a) tension test for dog-bone specimen, (b) compression test for specimens with VSF = 60% and 100%.

The displacement of 16 points was recorded for calculating the Poisson's ratio in this work [45]. As can be seen in Fig. 5, 16 red points were marked in the rotation centres for calculating the Poisson's ratio using the image processing method. The computing equations of the equivalent Poisson's ratio is shown in formulas (3) and (4):

Fig. 8

$$\nu_{ij} = -\frac{\epsilon_{ij}^x}{\epsilon_{ij}^y} = -\frac{\Delta_{ij}^x D_y}{\Delta_{ij}^y D_x} = -\frac{[X_{i(j+1)} - X_{i(j-1)}] D_y}{[Y_{(i+1)j} - Y_{(i-1)j}] D_x} \quad (2 \leq i \leq 3, 2 \leq j \leq 3), \quad (3)$$

$$\bar{\nu} = \frac{1}{4} \sum \nu_{ij} \quad (2 \leq i \leq 3, 2 \leq j \leq 3), \quad (4)$$

where D_x , D_y are the horizontal and vertical distance of P_{11} , P_{13} and P_{11} , P_{31} of undeformed structure, respectively (Fig. 5). i and j are the corresponding labels of the points in Fig. 5. X , Y are the real-time coordinates of these points.

4. Experimental results and discussion

The nominal stress vs nominal strain curves of two specimens are shown in Fig. 6(a), where the orange and blue curves are given for distinguishing VSF = 60% and VSF = 100%, respectively. And the numerical results are exhibited using the dotted curves. Similar to our previous works [47], three stages could be divided in each curve, including elastic stage, rib buckling stage, and enhanced stiffness stage. As shown in Fig. 6(b), the red-marked yield point and densification points are significant boundaries to three stages. k_1 , k_2 , k_3 , represent the slopes of the structure in different stages. The k_1 , k_2 , k_3 , ϵ_y , ϵ_d , α , and actual VSF of experiment and simulation have been given in Table 2, where α is stiffness ratio defined as formula (5). Actual VSF is the ratio of densification strains for the designed model and original VSF=100% model, which is defined for estimating the accuracy of variable stiffness design. From Table 2, k_2 of the two specimens exhibit a negative stiffness stage due to the rib buckling. The two specimens with VSF of 60% and 100% have a similar mechanical response before densification, but different equivalent modulus k_3 after densification in Fig. 6(a). The k_3 and α of VSF = 60% are obviously higher than 100%. This is the result of adding more materials to the design region for VSF = 60%, which also causes the compacted point to occur ahead of schedule as designed. Although some differences show in equivalent stiffness of experiment and finite element results, the overall trend and critical strain of nominal stress vs nominal strain curve are similar in Fig. 6(a). The deformation modes of simulation and test show a great agreement as well and a

detailed discussion of this part is conducted in the next section. As for producing differences shown in the results, the heat-affected edge effect caused by laser cutting may be the probable reason.

$$\alpha = k_3/k_2 \quad (5)$$

The onset of densification strain of cellular structures could be calculated using energy efficiency [71]. But the design of the compaction point in the VSF method is the complete densification strain, and our study has indicated that the actual VSF calculated by the initial densification strain is not accurate with design [47]. Thus, the complete densification strain is adopted as the compaction point in the following sections. The complete densification strain is defined as the turning position between the stage of enhanced slope and the fixed slope stage (Fig. 6(b)).

The evolutions of Poisson's ratio are calculated by the method of Section 3.3 (Fig. 7). It can be seen that the maximum negative Poisson's ratios of specimens with VSF = 60% and 100% are -0.65 and -0.84, respectively. And the corresponding nominal strains are about 0.09 and 0.15, respectively, which is consistent with the densification strain in the nominal stress-nominal strain curve. A considerable large error bar shows in Fig. 7 before the nominal strain increases to about 0.07 because the increase of denominator (axial strain) for calculating Poisson's ratio is smaller than that of the numerator (lateral strain). The deformation processes for two metallic samples with VSF=60% and VSF=100% are illustrated in Fig. 8. Similar to the results of Poisson's ratio vs nominal strain curves, two samples exhibit the contraction effect, and the phenomenon for the specimen with VSF=100% is more pronounced than VSF=60%. The FE deformation mode matches well with the experiment. From the stress distribution in FE graphs, the stress is mainly distributed in the joint of unit cells before densification. As the development of densification, the whole of unit cells gradually turned into a plastic stage, leading to the appearance of an enhanced stiffness stage.

5. Parametric study

In this section, firstly, the accuracy for this metallic auxetic meta-material between designed VSF and actual VSF is analyzed; secondly, the effect of compression velocity on tunable energy absorption is discussed in detail; lastly, a functionally graded auxetic structure is proposed and tested to reduce the initial peak force, and stiffness ratio in the densification stage.

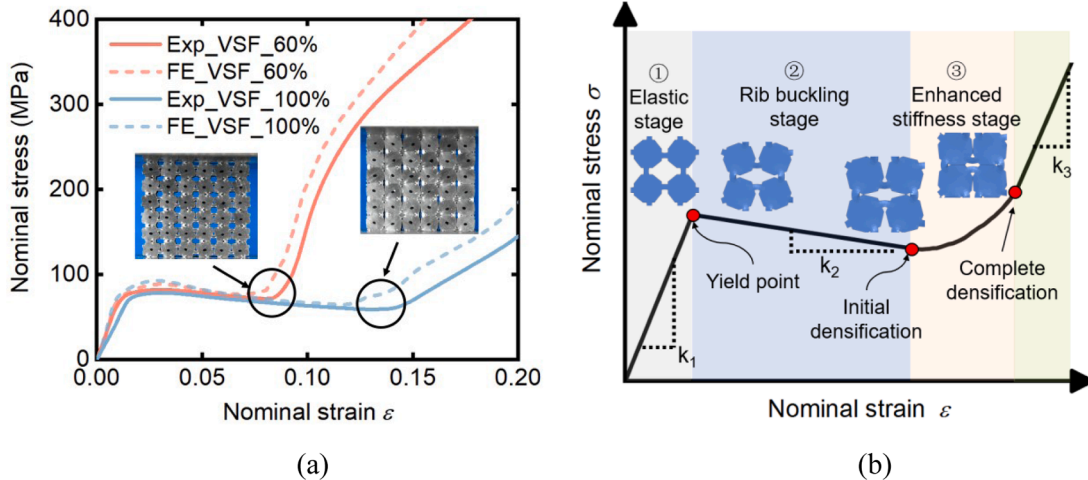
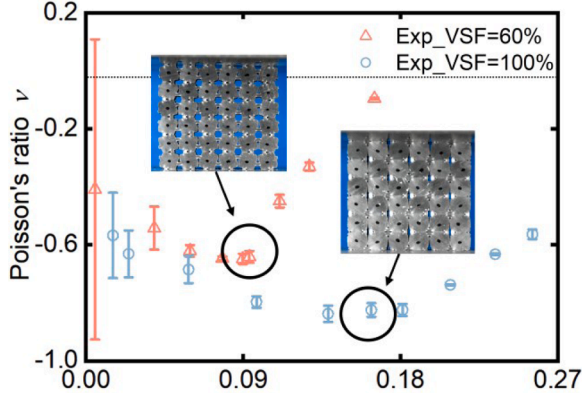


Fig. 6. Curves of stress vs strain for experimental specimens and ideal model: (a) experimental specimens with VSF=60% and VSF = 100%, (b) ideal stress vs strain curves of auxetic materials.

Table 2

Critical parameters of experimental stress vs strain curves.

	k_1 (MPa)	k_2 (MPa)	k_3 (MPa)	α	ε_y	ε_d	Actual VSF
Exp_VSF=60%	6918.3	-250.7	7977.8	31.82	0.010	0.089	61.38%
FE_VSF=60%	7648.2	-401.8	8794.3	21.89	0.011	0.086	60.99%
Exp_VSF=100%	4747.8	-202.1	1458.2	7.22	0.015	0.145	
FE_VSF=100%	7578.2	-335.8	1393.3	4.15	0.0095	0.141	

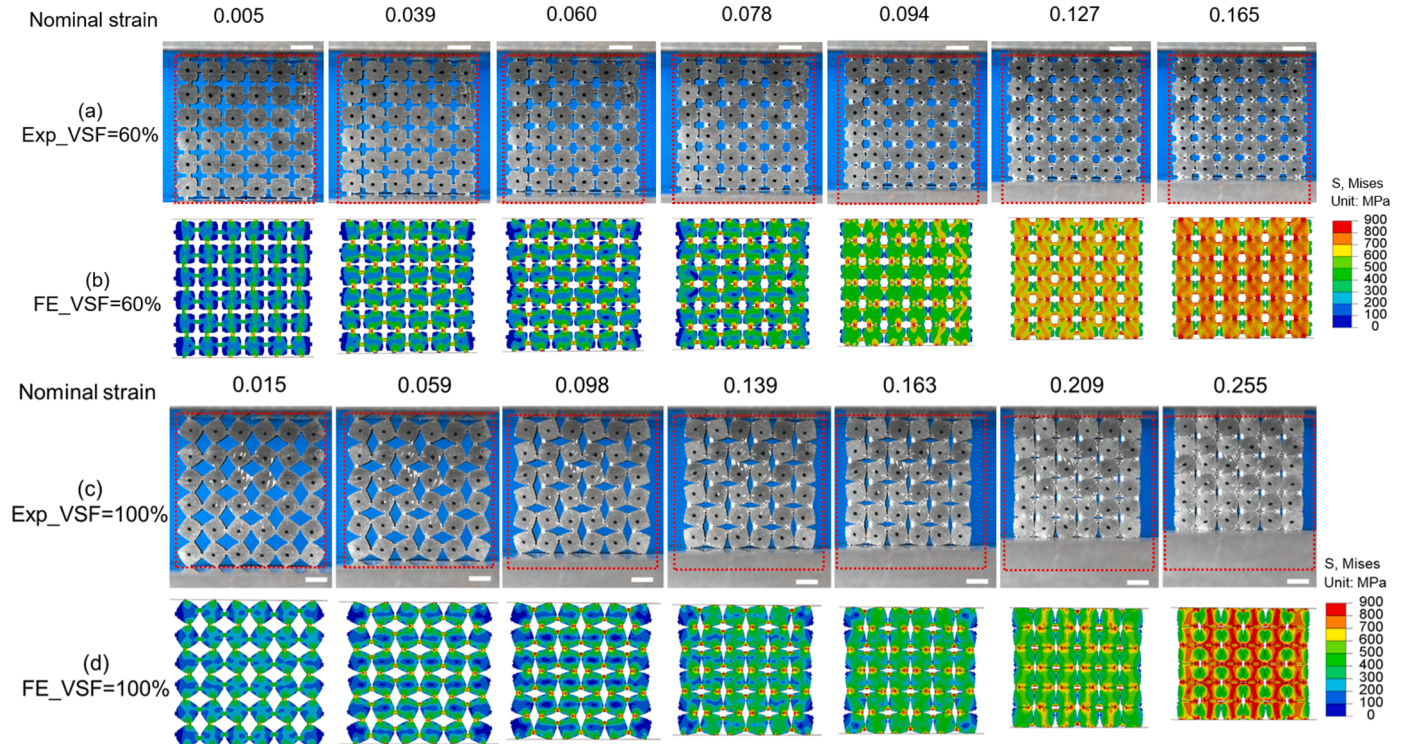
**Fig. 7.** The evolution of Poisson's ratio vs nominal strain.

5.1. Accuracy of designed VSF

The nominal stress vs nominal strain and Poisson's ratio vs nominal strain curves of four models with VSF = 40%, 60%, 80%, 100% are exhibited in Fig. 9. The densification points are highlighted by stars and the corresponding compacted unit cells are given in the figures. Similar to our previous work [47], these curves exhibit the phenomenon of tunable compacted points as expected. For examining the effect of base material on the designed structures, a comparison between metal and rubber material is conducted. The evolution of Poisson's ratio and the

deformation processes of the metallic specimens are in substantial agreement with rubber. However, many differences exist in the stress vs strain curves between this metallic specimen and the previous rubber specimen. The metallic specimen has a short elastic stage and obvious yield point due to the elastoplastic property. In addition, the stress would not increase when the stainless steel yield, which results in the rib buckling stage exhibits negative stiffness performance. The overall bearing capacity of metallic samples is about 500 times of rubber materials. Thus, metallic auxetic materials have wider application prospects in practical engineering. However, the metallic auxetic metamaterials with variable stiffness could only be used once under large deformation because of the non-recoverability of plastic deformation. Rubber-like materials are more appropriate to apply in the scenario with low and repeated load capacity.

The minimum Poisson's ratio ν_{min} , compacted point ε_d , and actual VSF are obtained by the curves, as shown in Table 3. The actual VSF percentages defined in the first paragraph of Section 4 are approximate to the designed. As the designed value of VSF increases, the auxetic effect also becomes larger, and the minimum Poisson's ratio is about -0.774. The specific energy absorption (SEA) calculated using Eq. (6) is given in Table 3, which is a significant indicator in the design of the absorber [72]. Because the densification strain is designed by adding materials in the design region quantitatively, the weight of the structures would be increased with changing of VSF from 100% to 0%. Besides, the compaction point ε_d would be increased monotonically as the designed VSF increase. Therefore, as the increase of VSF value, the SEAs are increased.

**Fig. 8.** Deformation processes of experiment and FE model (Scale bar: 10 mm).

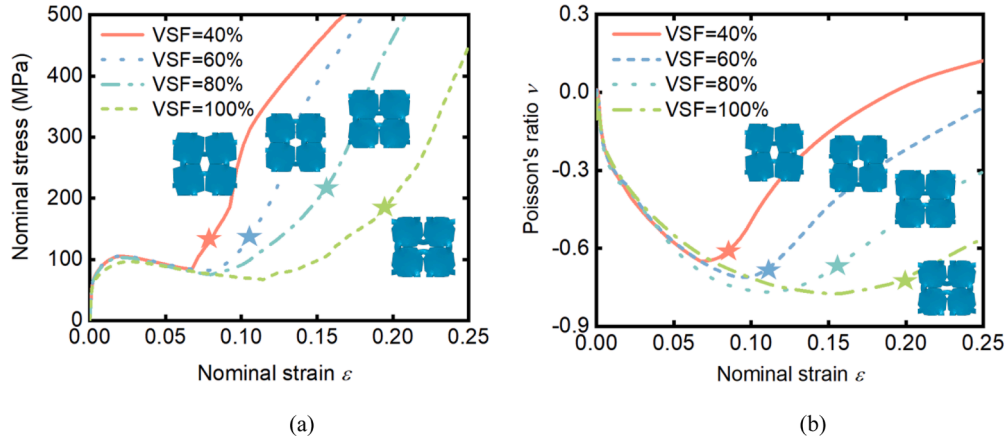


Fig. 9. Nominal stress vs nominal strain and Poisson's ratio vs nominal strain curves for four models with VSF = 40%, 60%, 80%, 100% (Star marks: densification points): (a) nominal stress vs nominal strain curves, (b) Poisson's ratio vs nominal strain curves.

Table 3

Comparison of the designed and the actual VSFs of the models under static load.

	VSF=40%	VSF=60%	VSF=80%	VSF=100%
ν_{min}	-0.649	-0.711	-0.76	-0.774
SEA (J/g)	1.28	2.01	3.43	4.03
ε_d	0.0797	0.1146	0.1620	0.1991
Actual VSF	40.03%	57.56%	81.37%	1

$$SEA = \frac{EA}{m} \quad (6)$$

Where EA is the envelope size integrated from $\varepsilon = 0$ to ε_d in load-displacement curve. m is the mass.

Before densification, the increment between the envelope area and nominal strain for stress vs strain curves with different VSF (Fig. 9(a)) would exhibit linear relation due to the approximate stress value and trend, which means the variable energy absorption factor (VEF) inspired by the conception of variable stiffness factor (VSF) is feasible and robust. VEF is defined as formula (7), which is desirable to evaluate the accuracy of tunable energy absorption under uniaxial crushing with different velocities.

$$VEF = \frac{E_1}{E} \quad (7)$$

Where E_1 is the absorbed energy of model with designed VSF, E is the absorbed energy of model with VSF=100%. Absorbed energy is calculated by the strain integration from 0 to ε_d .

5.2. Crushing velocity

In the dynamic simulation, the mesh types and contact are the same as the abovementioned quasi-static compression. The impact load is applied by a rigid face sheet subjected to a constant crushing velocity. And the other rigid sheet is fixed in the axial direction. The boundary and load conditions are similar to these publications [65,66]. The 304 stainless steel is a sensitive material to strain ratio and temperature, which results in the elastic-plastic model is unusable in crushing load. According to the dog-bone test in Section 3.3, the material properties are like the literature [73] in quasi-static. Consequently, the same materials parameters are adopted in the studies of crushing velocity.

The Johnson-Cook model is a common constitutive model to reflect the dynamic plastic behavior in finite element analysis [74,75]. The Johnson-Cook parameters of 304 stainless steel are determined by fitting the test curves in different loading strain rates [73], as shown in Table 4. According to the publication [73], the FE model with the same geometrical parameters is established to verify the validity of the impact

Table 4

The Johnson-Cook parameters of 304 stainless steel [73].

Density (kg/ mm ³)	Young's Modulus (GPa)	Poisson's ratio	Melting temperature (K)	Room temperature (K)	Specific heat (J/ kg K)
7900	200	0.3	1673	293	440
A (MPa)	B (MPa)	n	c	$\dot{\varepsilon}_0$ (1/s)	m
310	1000	0.65	0.07	1.00	1.00

simulation. The comparison of the Kolsky Bar experiment, FE in publication [73] and FE result in our study are shown in Fig. 10. The orange curve represents the nominal stress vs nominal strain curve in our study, which matches well with the FE and experiment results in the publication [73]. Besides, the deformation modes under 550 1/s strain rate are compared in Fig. 11. The results also indicated the validity of the FE model.

The same FE settings with the verified model are utilized in crushing velocity analyses. The effect of velocity on the accuracy of the VSF method is studied numerically. Five groups of velocities including 5 m/s (79 1/s), 25 m/s (397 1/s), 50 m/s (793 1/s), 75 m/s (1190 1/s), and 100 m/s (1587 1/s) are simulated. Due to the jittery stress vs strain curves in the dynamic simulation, the densification point is mainly determined by the Poisson's ratio vs nominal strain curve and deformation mode. As shown in Fig. 12, the compacted unit cells are marked using red frames. By combing the variation trend of Poisson's ratio, the densification points are determined and highlighted by stars in Fig. 12.

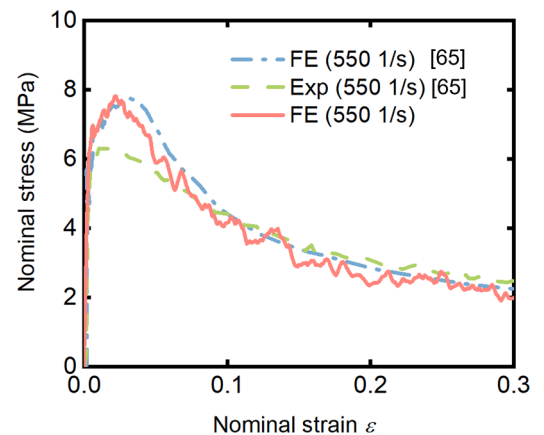


Fig. 10. The comparison of stress vs strain curves with loading (550 1/s) between this paper and the publication [73].

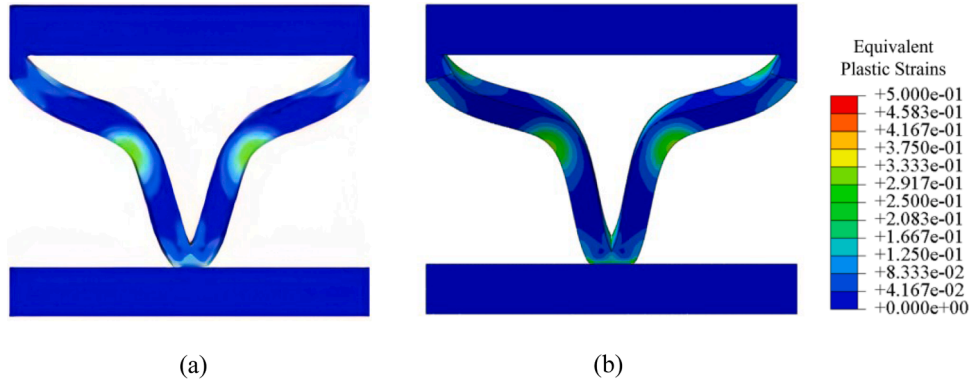


Fig. 11. The comparison of deformation between the publication and this paper under 0.17 strain: (a) the publication [73], (b) this paper.

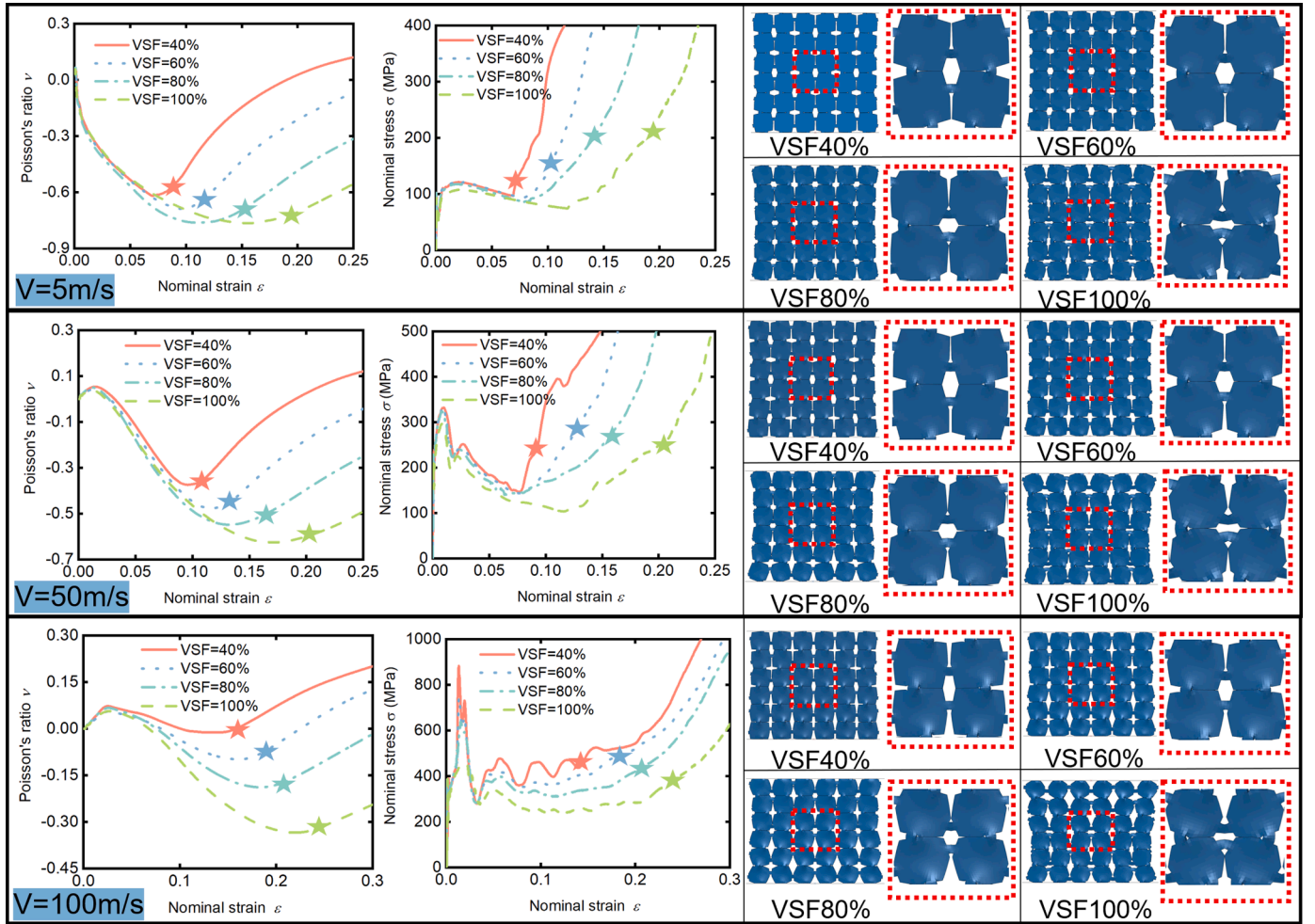


Fig. 12. Effects of crushing velocity on Poisson's ratio, stress vs strain response, and densification strain for four designed models (Star marks: densification points).

The mechanical responses of $V = 25$ m/s and 75 m/s have similar characteristics with velocities of $V = 50$ m/s and 100 m/s, respectively. Hence, as representative crushing velocities, the results of FE models under $V = 5$ m/s (low), 50 m/s (medium), and 100 m/s (high) are exhibited in Figs. 12 and 13, i.e., nominal stress vs nominal strain, Poisson's ratio vs nominal strain curves and deformation processes of $VSF = 40\%$.

The auxetic behavior and densification strains are synchronously improved as the designed VSF increases because the trough of Poisson's ratio curve is approximate to the compacted point. As the crushing

velocity increases, the negative Poisson's ratio effect is diminished. Especially in a high-velocity of 100 m/s, the minimum Poisson's ratio has turned positive when the VSF is designed as 40% . In the initial impact when $V = 50$ m/s and 100 m/s (before 0.01 strain), the Poisson's ratio of models with different VSF firstly increases from 0 to positive. From the deformation processes in Fig. 13, the abovementioned results are mainly caused by the features of shock fronts where the rotation of unit cells occurred locally. Significantly, due to the enhanced shock wave effect of cellular in high-impact velocity [76], the unit cells in shock fronts are compacted under 0.058 compression strain, but the unit

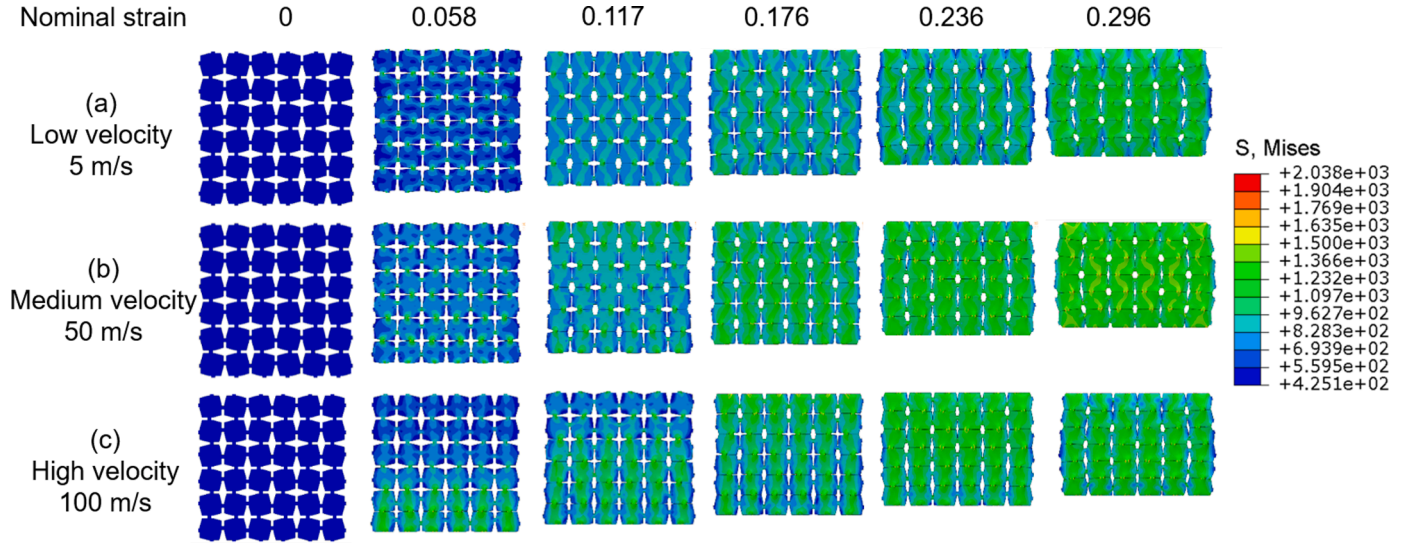


Fig. 13. Deformation for models with VSF=40% under different crushing loads.

cells adjacent to the fixed end have no obvious deformation under $V = 100$ m/s (Fig. 13(c)). For this reason, the fluctuation of nominal stress vs nominal strain curves gradually improved with the impact velocity increased. An evident stiffness-enhanced stage could be observed (marked by stars) under low and medium velocity. While the stars in curves of $v = 100$ m/s could rarely represent the stiffness-enhanced performance, which indicates that the VSF design methodology might be invalid to high-velocity crushing.

To further verify the agreement between the designed VSF and actual VSF, the actual VSFs are calculated according to the curves in Fig. 12, as shown in Fig. 14. The designed VSFs of 40%, 60%, 80% are given and marked by red color. The results illustrate that the actual VSFs are approximate with designed ones under low and medium velocities, including 5 m/s, 25 m/s, 50 m/s. With the crushing velocity increases to 75 m/s and 100 m/s, the actual VSF is rather larger than the designed ones. The actual variable energy factor (VEF) defined in Section 5.1 is given for verifying the accuracy of tunable energy absorption. Similar to actual VSF, the actual VEFs are slightly higher than designed values under velocities of 5 m/s and 25 m/s. When the impact increases to high-velocity (over 50 m/s), the actual VEF sharply increases and is much larger than the designed value. These results show that the tunable stiffness and energy absorption are only effective under low and medium impact. By carefully analyzing the curves and deformation in Figs. 12 and 13, the reasons for the results could be concluded as the characteristic of shock fronts. Especially in high velocity, the plastic deformation of the material occurred before the elastic rotation behavior of auxetic unit cells, which results in the complete densification strain increased. Meanwhile, from the curves of $V = 100$ m/s in Fig. 12, the marked densification strains are obviously delayed in comparison with other velocities. The real VEFs have the same trend with real VSF, but the percentage is slightly higher than real VSF. The reason is that the strain ranges of the enhanced stiffness stage are extended with the increase of designed VSF (from 40% to 100%). The envelope area of force vs displacement (energy absorption) exceeded the designed value due to the increase of reaction forces in the enhanced stiffness stage.

The SEAs of models with different VSF values are shown in Fig. 15. With the increase of the VSF percentage, the SEA is increased gradually, which is similar to the results in quasi-static load. Likewise, the SEA would be improved with the increase of crushing load. When the crushing load increased to more than medium velocity (50 m/s), the increment of SEA is obviously higher than that of low velocities (5 m/s and 25 m/s). This phenomenon also proved the different mechanical responses of the structures under low and high crushing loads.

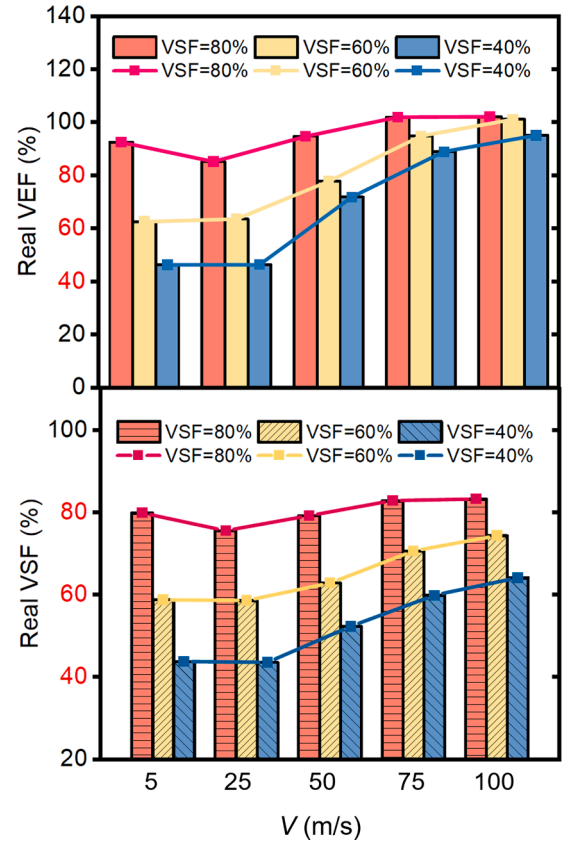


Fig. 14. The comparison of the accuracy between the actual VSF, VEF (height of the column) and designed VSF (highlighted by red color) under different crushing loads.

5.3. Discussion of functionally graded auxetic structure

The stiffness ratio α and the initial peak force P_{max} defined in Section 4 are two primary indicators to the design of anti-impact equipment. The high peak force and sharp increase of equivalent stiffness in nominal stress vs nominal strain curve manifest as the increased P_{max} and α , which is unfavorable to achieve high energy absorption. However, the results of Section 5.2 show that the models with VSF=40% have a

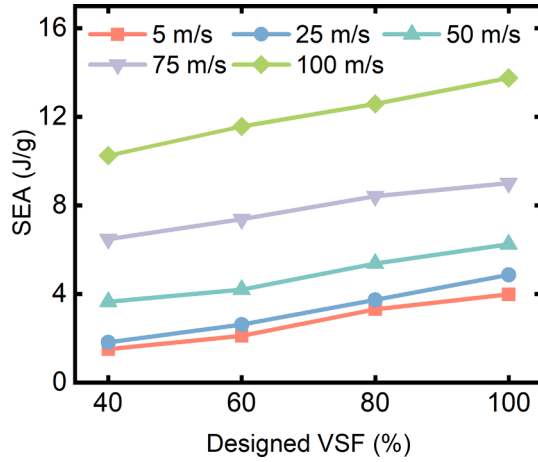


Fig. 15. The SEA of models with different VSF values and under different crushing loads.

considerably larger stiffness ratio of 8.75 than that of 5.81 in VSF=100% under impact velocity of 25 m/s. The sharp increase of nominal stress occurs in the densification point under low and medium velocities for the model with VSF = 40% (Fig. 12). Thus, the layered gradient (LG) structure is proposed to avoid the excessive sharp increase of stress and to decrease the initial peak force.

The overall geometric size of the specimen is the same as the designed one in Section 3. Three layers including VSF = 30%, 60%, and 100% are divided along with the height direction, as shown in Fig. 16. The nominal stress vs nominal strain response of quasi-static compression is divided into 5 stages by critical strains. (Fig. 16①~⑤). The deformation modes in the five nominal strains are shown in Fig. 17. The first stage is elastic and the deformation is observed hardly (Fig. 17①). When the compression strain reaches to about 0.073, the bottom layer of VSF = 30% is compacted (Fig. 17②), which results in the equivalent stiffness firstly increased (Fig. 16②). Then, the layer of VSF = 60% becomes densified, and the layer of VSF=100% just starts to densify (Fig. 17③). A short stress plateau is appeared (between ③ and ④ in Fig. 16) until the strain is about 0.107 due to the cellular rotation in the layer of VSF = 100% (Fig. 17④). The design region of those unit cells contacts completely, resulting in the whole layer gradient auxetic structure reaches final densification (Fig. 17⑤). The nominal stress vs nominal strain curve and deformation of the verified FE model are given to compare with the experiment. The results indicate that the above-mentioned discussions are reliable under the verifying of great agreement between the trend of mechanical curves and deformation modes in

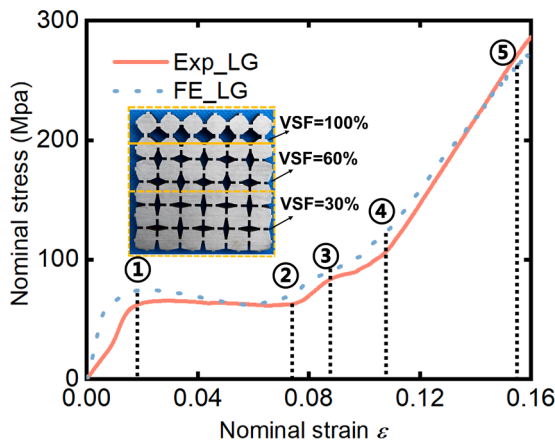


Fig. 16. Experimental and FE nominal stress vs nominal strain curve for layered gradient auxetic structure.

the experiment and FE model. A difference occurred in the elastic stage and the stress of FE is slightly higher than the test data. This phenomenon, by carefully checking, is brought by the manufacturing defect during the cutting process.

The stiffness ratio significantly decreases due to the layered densification under quasi-static compression. To examine the results of layered gradient (LG) auxetic structures under the crushing load, the models are analyzed numerically under constant crushing loads of 5 m/s (79 1/s), 25 m/s (397 1/s), 50 m/s (793 1/s), 75 m/s (1190 1/s), and 100 m/s (1587 1/s). The setups of the FE model are the same as in Section 5.2. The relations between nominal stress, Poisson's ratio and nominal strain of low (5 m/s), medium (50 m/s), and high velocities (100 m/s) are exhibited in Fig. 18. LG_30% and VSF_30% are named to represent the layered gradient (LG) and original auxetic structures designed by variable stiffness factor (VSF). It can be seen that the elastic and stress plateau stages of the two models are similar under 5 m/s and 50 m/s. The densification strain and Poisson's ratio of LG_30% is slightly larger than VSF_30%. In the high-speed crushing of 100 m/s, the peak force is obviously decreased using the gradient design. Some critical parameters obtained from Fig. 18 are given in Table 5. It is worth noting that the stiffness ratio of LG_30% is significantly lower than that of VSF_30% in static and dynamic load. The peak force of the layered gradient structure exhibits a lower initial peak force than the original VSF structure under a medium crushing load. And the gap of peak force between the LG_30% and VSF_30% would be increased with the crushing velocity increased. Besides, the SEAs are calculated to assess the crashworthiness of layered gradient structure in Table 5, where the LG_30% has a higher SEA than VSF_30% in the same crushing load. The aforementioned results indicate that the LG_30% possesses better energy absorption performance than VSF_30%. The mechanism of such gradient structure is that the compaction process is prolonged by layered densification, and the front part of the structure holds 100% VSF, where a high void ratio provides the decreased peak force.

6. Concluding remarks

In this study, a series of auxetic metamaterials were proposed using the variable stiffness factor (VSF) method. Two 2D metallic specimens were designed, manufactured and tested to examine mechanical responses under uniaxial compression. The validated metallic models were established numerically to verify the accuracy of the VSF method under quasi-static and impact load. To enable practical applications of the VSF method in impact-resistant design, the variable energy factor (VEF) was introduced to achieve tunable energy absorption. In addition, a novel functionally graded auxetic structure based on the VSF method was investigated experimentally and numerically, aiming to decrease the stiffness ratio and peak force under impact load. From this study, the following conclusions can be drawn:

- (1) The densification strain of auxetic metamaterials designed using the VSF method could be tuned accurately, and such a property could be achieved for both rubber-like materials and metallic materials.
- (2) The proposed variable energy factor (VEF) could effectively evaluate the accuracy of tunable energy absorption. Under low- and medium-crushing loads, the designed structure had an actual VEF percentage close to the designed value.
- (3) In the test of low- and medium-velocity impacts, the peak force and stiffness ratio of improved functionally graded auxetic structure could be decreased, which is desirable in impact-resistant design. Besides, the densification strain was slightly larger than the original VSF designed structures.
- (4) As the impact velocity increased from low and medium to high, the Poisson's ratio gradually changed from negative to positive. Likewise, the actual VSF and VEF percentages were higher than the designed value under high-velocity impact.

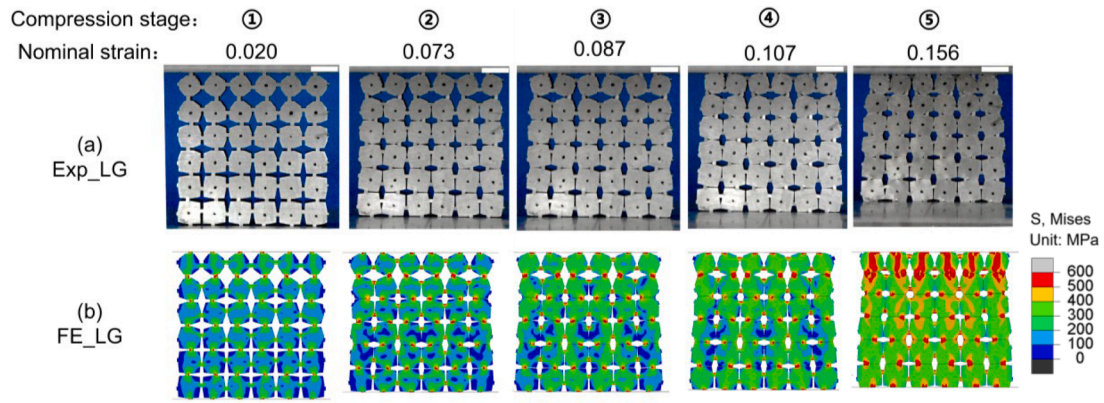


Fig. 17. Deformation processes of functionally graded auxetic structure (Scale bar: 10 mm).

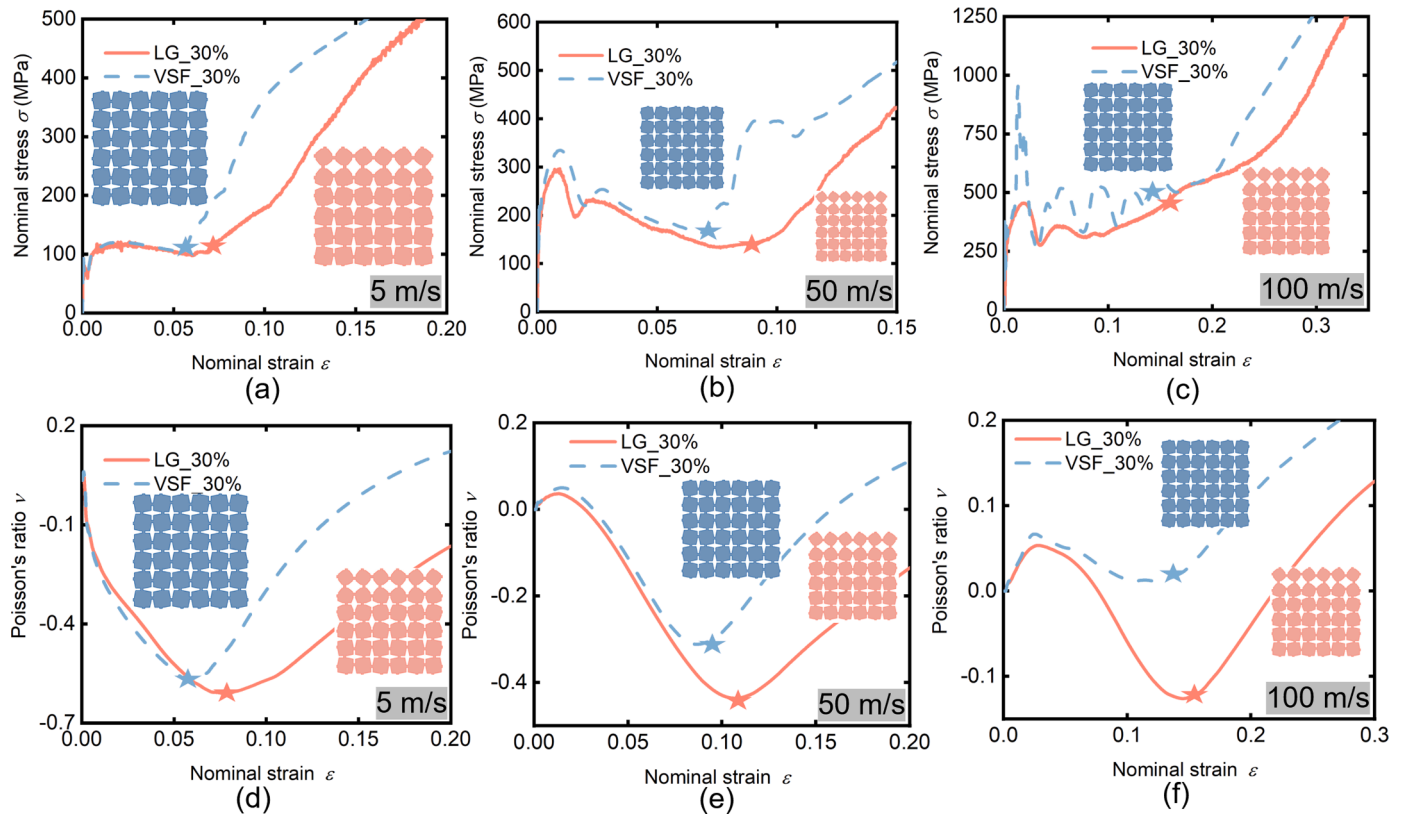


Fig. 18. Numerical results for the two designed models: (a) the σ vs ϵ curves under 5 m/s, (b) the σ vs ϵ curves under 50 m/s, (c) the σ vs ϵ curves under 100 m/s, (d) the ν vs ϵ curves under 5 m/s, (e) the ν vs ϵ curves under 50 m/s, (f) the ν vs ϵ curves under 100 m/s.

Table 5

Primary parameters of VSF_30% and LG_30% under different crushing velocities.

Velocity (m/s)	LG_30%				VSF_30%			
	α	P_{max} (MPa)	ν_{min}	SEA (J/g)	α	P_{max} (MPa)	ν_{min}	SEA (J/g)
0	5.45	87	-0.65	1.1	9.28	89	-0.60	0.86
5	3.68	121	-0.61	1.6	12.57	120	-0.58	1.2
25	7.91	202	-0.55	2.3	25.30	230	-0.49	1.8
50	2.71	297	-0.44	3.6	5.22	336	-0.31	3.1
75	1.80	393	-0.28	6.5	2.30	544	-0.09	5.5
100	2.87	455	-0.13	10.2	7.37	953	0.01	9.0

The energy absorption capacity of impact-resistant equipment could be designed quantitatively using the VSF method. Depending on the application scenarios, different matrix materials could be selected to fabricate these auxetic metamaterials, i.e., polymer and metal with

varying degrees of Young's modulus. However, this design method also has some limitations, such as the maximum designed energy absorption should be less than the structure of VSF = 100% and the base material should have enough elastic or plastic deformation properties. In

addition, the 3D auxetic metamaterials composed of unit cells with variable stiffness, including tubes and cubes, are also worth further investigating. These auxetic structures with different geometric forms would broaden the potential applications in the fields of civil engineering, vehicle engineering and protective engineering.

CRediT authorship contribution statement

Yi Zhang: Conceptualization, Methodology, Software, Writing – review & editing. **Xin Ren:** Project administration, Supervision, Funding acquisition, Writing – review & editing. **Dong Han:** Investigation, Writing – review & editing. **Xian Cheng:** Investigation. **Wei Jiang:** Writing – review & editing. **Xue Gang Zhang:** Resources. **Xiang Yu Zhang:** Methodology, Writing – review & editing. **Yi Min Xie:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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